

Pipeline Notes

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The Big Picture

The pipeline answers one question: **how much more concentrated is a protein inside a condensate compared to the surrounding nucleus?** That ratio is the Partition Coefficient (PC). To compute it you need to know *where* the condensates are and *where* the nucleus is — that's what the pipeline figures out automatically.

Step-by-Step

1. Load

Two 3D image stacks are read in from TIF files:

- **Ch1** — the nuclei channel (fluorescent signal from the nucleus)
- **Ch2** — the condensate channel (fluorescent signal from the protein of interest)

Each stack is $55 \text{ Z-slices} \times 185 \times 259$ pixels. Think of it as a 3D block of pixels, like a loaf of bread sliced 55 times.

2. Denoise

Before trying to find any boundaries, every Z-slice gets run through Cellpose's denoising model. This sharpens edges and suppresses shot noise so the segmentation model can draw tighter, more accurate boundaries.

Key point: the denoised image is thrown away after this step. It is only used to guide *where* the boundaries go — never to measure intensity values.

3. Segment

Cellpose 3 (cyto3 model, `do_3D=True`) processes the full 3D stack — not slice by slice, but all 55 slices simultaneously. This means it tracks the same object consistently across Z rather than redrawing boundaries independently on each slice.

The output is two sets of **masks**:

- **Condensate masks** — each condensate gets a unique integer label. Every pixel inside condensate #1 is labeled 1, condensate #2 is labeled 2, and so on.
- **Nuclei masks** — same idea for nuclei.

After segmentation, the nuclei masks go through one extra step: Cellpose tends to split each nucleus into 15–25 fragments because of internal texture (the condensates inside the nucleus confuse it). To fix this, the binary mask (any pixel labeled > 0 = “inside a nucleus”) is run through 3D connected-component analysis. This collapses all the fragments back into the true connected regions — 76 Cellpose labels become 5 clean nuclei.

4. Measure

Using the masks as selection tools on the original raw image, **regionprops** extracts per-object measurements for every Z-slice: area, centroid, and mean intensity.

5. 3D Volume

Counts how many voxels (3D pixels) each labeled object occupies across all 55 slices. This gives a true 3D volume estimate for each condensate and nucleus rather than just a per-slice area.

6. Partition Coefficient

This is where the masks are used to select pixels from the **raw image** (Ch2, the condensate channel).

Background (B)

Find the minimum pixel value across the entire 3D stack. This is the camera/autofluorescence floor — a baseline offset that has nothing to do with the protein. It gets subtracted from every pixel before any density is computed.

Condensate density

- Take all pixels inside *both* a condensate mask *and* a nucleus mask
- Subtract B from each, clip to 0
- Sort by intensity, keep only the top 75% brightest
- Take the mean

The top-75% trim is important: Cellpose draws slightly inflated boundaries that include a dim halo around the true bright core. Trimming the dimmest 25% of mask pixels recovers approximately what a researcher would have drawn by hand.

Dilute density

- Take all pixels inside a nucleus mask but *outside* any condensate mask — this is the quiet nuclear background
- Find all valid $10 \times 10 \times 10$ voxel patches that fit entirely within this region ($\sim 88,000$ patches)
- Sort by mean intensity, take the 50 lowest-intensity patches
- Average them

The lowest-50-patch approach mimics how a researcher manually picks a quiet, representative background spot. Using a single random patch was unstable (PC would swing from 4.3 to 6.0 depending on where it landed).

PC formula

$$\text{PC} = \text{condensate_density} / \text{dilute_density}$$

Interpretation: how many times brighter are condensates than the surrounding nuclear background, after removing the camera offset.

Why Each Step Matters

Step	What would break without it
Denoising	Blurry boundaries → masks include too much dim halo → PC underestimated
do_3D=True	Slice-by-slice segmentation gives inconsistent boundaries across Z → noisy intensity measurements
CC nuclei fix	Fragmented nucleus labels leave gaps → condensates in gaps excluded from PC
Background subtraction	PC systematically lower than reference (reference always subtracts B)
Top-75% trim	Cellpose halos drag cond_density down → PC underestimated vs. manual reference
Lowest-50-patch dilute	Single random patch is seed-dependent → PC swings $\pm 30\%$ across runs

What PC Means Biologically

A PC of 1.0 means the protein is equally distributed inside condensates and in the surrounding nucleus — no enrichment. A PC of 6.3 (our result, matching the manual reference) means the protein is $\sim 6\times$ more concentrated inside the condensate than in the nuclear background. Higher PC = tighter, more selective condensate.

Batch Validation Metrics Explained

The batch run across 30 JABr cells reports three numbers: r , RMSE, and bias.

Pearson r (correlation)

Measures how well the pipeline's PC values *track* the reference values — do they go up and down together? $r = 1.0$ is perfect, $r = 0$ is no relationship. $r = 0.735$ means the pipeline ranks cells in roughly the right order (high PC cells are still high, low PC cells are still low), but it is not a tight 1:1 relationship.

RMSE (Root Mean Square Error)

The average size of the error per cell, in the same units as PC. RMSE= 2.815 means on average the pipeline is off by ~ 2.8 PC units per cell. Since the reference PCs range from ~ 4 to ~ 20 , that is a moderate error — good for most cells, worse for the outliers.

Bias (+2.8%)

The systematic over- or under-estimation averaged across all cells. +2.8% means the pipeline is very slightly over-estimating PC on average — essentially zero systematic error. A large positive bias would mean the pipeline always reads high; a large negative bias would mean it always reads low. The top-75% trim was specifically tuned to get this close to zero (it was -16% before the fix).

Metric	Value	What it tells you
Pearson r	0.735	Does it correlate? (shape of relationship)
RMSE	2.815 PC units	How far off is it on average? (absolute error)
Bias	+2.8%	Does it consistently over/under-shoot? (systematic offset)